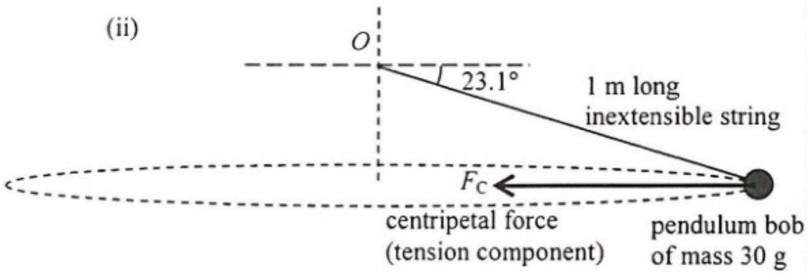
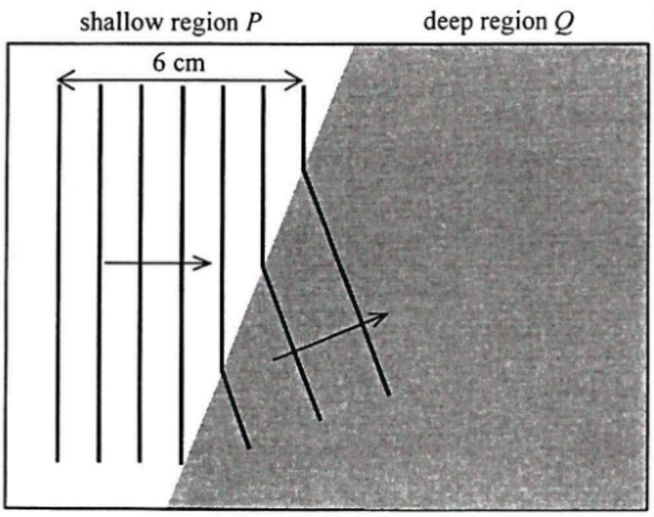
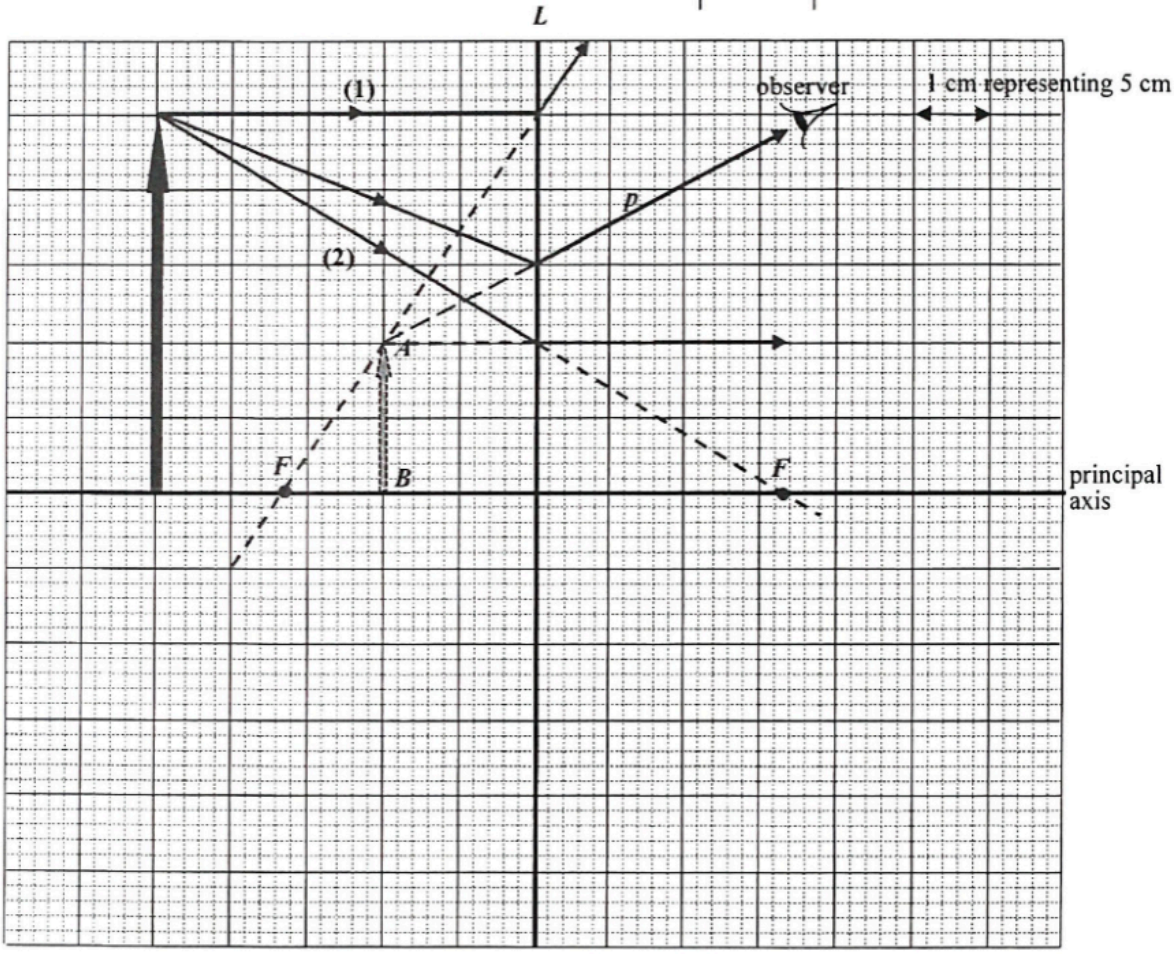
1. (a) (i)	$(1.5)(4200)(60-10) = m(3.34 \times 10^5) + m(4200)(10-0)$ $m = 0.837766 \text{ kg} \approx 0.838 \text{ kg}$		1M+1M 1A 3	
(ii)	More ice is needed to cool the container as it will release heat / thermal energy as well.		1A 1A 2	
(b) (i)	Conduction: - Foam is a poor conductor (of thermal energy) and it minimizes the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag). OR Radiation: - The shiny surface (inside) reduces the transfer of heat / thermal energy from the surroundings to the cool contents / ice cream (inside the bag) through emission of radiation OR Convection: - The zipper prevents convection between the hot air outside and the cool contents / ice cream (inside the bag).	Any **ONE**	1A 1	Note : Convection is the least effective transfer process in this case as the hot air outside is above the denser cool air around the ice cream.
(ii)	(Radiation) Make the outer surface (of the bag) shiny.		1A 1	

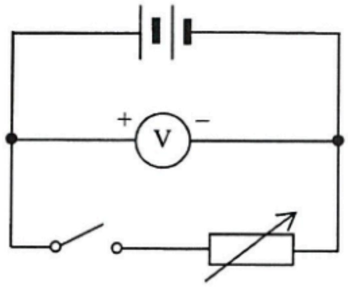
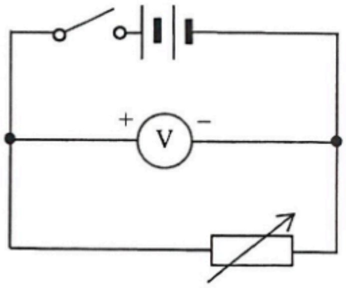
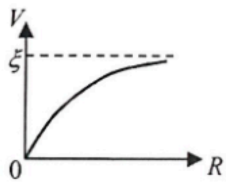
2. (a)	$pV = nRT$ $(100 \times 10^3)(0.52) = n\,(8.31)(273+15)$ $n = 21.727504 \text{ (mol)} \approx 21.7 \text{ (mol)}$	1M 1A 2	Accept: (21.0 ~ 22.0) mol
(b) (i)	Since $pV = nRT \Rightarrow V = \dfrac{nRT}{p}$ / volume V of the balloon depends on both T and p, the (fractional) decrease in pressure p (with height) is greater / faster than the (fractional) decrease in temperature T.	1A 1A 2	Accept: volume V decreases as temperature T decreases would be true only if pressure p is constant, however, p decreases with height.
(ii) (1)	$\dfrac{pV}{T} = \text{constant}$ $\dfrac{(100)(0.52)}{(273+15)} = \dfrac{p(8)}{216}$ $p = 4.875 \text{ kPa or } 4875 \text{ Pa}$	1M 1A 2	
(2)	$p = p_0 e^{-kx}$ $4.875 = 100\,e^{-0.138\,x}$ $x = 21.89166726 \text{ (km)} \approx 21.9 \text{ (km)}$	1M 1A 2	Accept: (21.7 ~ 22.0) km

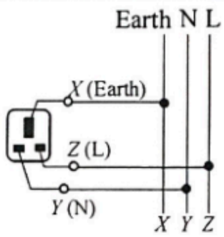
3. (a) (i) (1) $\frac{1}{2}mv^2 = mgh$	1M		
$v^2 = 2(9.81)(12)$			
$v = 15.344054 \text{ m s}^{-1} \approx 15.3 \text{ m s}^{-1}$	1A		Accept: $(15.0 \sim 15.5) \text{ m s}^{-1}$
($v = 15.491933 \text{ m s}^{-1} \approx 15.5 \text{ m s}^{-1}$ for $g = 10 \text{ m s}^{-2}$)		2	
(2) $s = \frac{1}{2}gt^2$	1M		
$12 = \frac{1}{2}(9.81)t^2$			
$t = 1.564124 \text{ s} \approx 1.56 \text{ s}$	1A		Accept: $(1.50 \sim 1.60)$ s
($t = 1.5491933 \text{ s} \approx 1.55 \text{ s}$ for $g = 10 \text{ m s}^{-2}$)		2	
(ii) $F - mg = \frac{mv - mu}{t}$			
$F = \frac{70 \times (0 - (-15.3))}{0.3} + 70 \times 9.81$	1M+1M		
$= 4266.9793 \text{ N} \approx 4270 \text{ N}$	1A		Accept: $(4180 \sim 4320)$ N
($F = 4314.7845 \text{ N} \approx 4310 \text{ N}$ for $g = 10 \text{ m s}^{-2}$)		3	
(iii) Elastic potential energy	1A	1	
(b) (i) (Velocity is too high, hence the force for deceleration is very large.) - The life net may be torn. - The falling person may be injured. - The firemen may not be able to hold the life net tight. Any **ONE**	1A	1	
(ii) There exists a horizontal velocity when a person jumps	1A		
and the horizontal displacement is very difficult to estimate as it depends on the time of fall, which is relatively long.	1A	2	

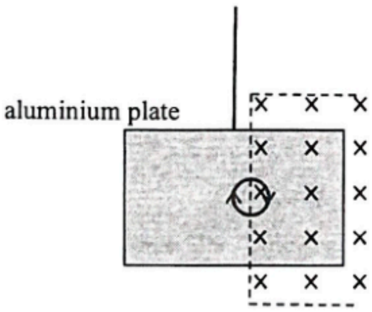
4. (a) (i)	Rotation rate $= \dfrac{\omega}{2\pi} = \dfrac{5.0}{2\pi}$ $= 0.795775$ (rev s^{-1}) ≈ 0.80 (rev s^{-1})	1M/1A	Accept: (0.79 ~ 0.80) rev s^{-1}
		1	
(ii)			
	F_C correctly indicated.	1A	
	$F_C = mr\omega^2$ $= (0.03)(1 \times \cos 23.1°)(5.0)^2$ $= 0.689866$ N ≈ 0.690 N ($F_C = 0.7033402$ N ≈ 0.703 N for $g = 10$ m s^{-2})	1M 1A	
	OR $T\cos\theta = F_C$ and $T\sin\theta = mg$ $F_C = \dfrac{mg}{\sin\theta}\cos\theta = 0.689866$ N ≈ 0.690 N	1M 1A	
		3	
(iii)	Horizontal component of tension provides the centripetal force, thus tension is larger than the centripetal force.	1M 1A	
	OR $T\cos\theta = F_C$ $\Rightarrow T > F_C$	1M 1A	
	OR $T\sin\theta = mg$ $\Rightarrow T\,(0.750\text{ N}) > F_C\,(0.690\text{ N})$	1M 1A	
		2	
(b) (i)	The gravitational force is always perpendicular to the moon's motion / displacement / velocity, thus no work is done on the moon by this force (k.e. unchanged)	1A 1A	
		2	
(ii)	(The claim is incorrect) as, by Newton's third law of motion, gravitational force of the same magnitude (but in opposite direction) is acting on the Earth by the moon.	1A 1A	Accept: action and reaction pair
		2	

5.	(a)	(i)	wavelength $\lambda = \dfrac{0.06}{7-1}$ $= 0.01$ m (= 1 cm)	1A	
				1	
		(ii)	speed $v = f\lambda = 10 \times 0.01$ $= 0.1$ m s^{-1} (= 10 cm s^{-1})	1M/1A	
				1	
	(b)	(i)	frequency = 10 Hz	1A	
				1	
		(ii)		1A 1A	
				2	
		(iii)	Refraction. It is due to the change in wavelengths / wave speeds in different media / depths.	1A 1A	
				2	

6.	(a)	*L* is diverging / concave. Only diverging / concave lens forms diminished, virtual image.	1A 1A 2	
	(b)	 Correct position and height of object	2A 2	
	(c)	Correct ray to locate *F* and focus *F* correctly marked. Focal length = 16.5 cm	2M 1A 3	Accept: (15.5 ~ 17.5) cm
	(d)	Correct ray *p* from tip of object	2A 2	

7. (a) 	1A 1A	Correct circuit with correct symbol Correct polarity Alternative circuit 
Close the switch and record corresponding V and R readings Adjust the resistance R to lower/other value(s) and repeat the experiment	1A 1A	
Precaution: - First set the variable resistor to its maximum / a large value - Open the switch after each measurement - Any reasonable answer } Any **ONE**	1A	
	5	
(b) Terminal voltage V delivered increases with increasing (loading) resistance R (or graphical representation)	1A	Accept: 
$V = \xi \dfrac{R}{R+r}$ <u>OR</u> $V = \xi - \dfrac{\xi}{R+r} r$	1A	
	2	

8. (a) mains socket 	1A	
	1	
(b) (i) - If one of the lighting sets / circuits fails, the other (in parallel) can still operate, i.e. both work independently. - Both can work at the rated power. - Any reasonable answer } Any **ONE**	1A	
	1	
(ii) $P = IV$ $(300 + 450) = I(220)$ $\boxed{I = \frac{P_1}{V} + \frac{P_2}{V} = \frac{300}{220} + \frac{450}{220}}$	1M	
$I = 3.409091\text{ A} \approx 3.41\text{ A}$	1A	
Thus 5 A fuse should be used.	1A	
	3	
(c) Electrical energy used per day $= 0.500\text{ kW} \times 8\text{ h} + 2\text{ kW} \times 0.5\text{ h} + 3\text{ kW} \times 2\text{ h}$ $= 11\text{ kW h}$	1M	
Cost $= \$0.9 / \text{kW h} \times 11\text{ kW h}$	1M	
$= \$9.9$	1A	
	3	

9.	(a)	(i)	By Lenz's law, an e.m.f. would be induced such that it opposes the change, i.e. decrease of magnetic flux (into the paper) by driving an induced current (clockwise) in the coil / circuit (complete).	1A 1A	
				2	
		(ii)	$N\Delta\Phi = NBA$ $= (20)(0.3)(0.005)$ $= 0.03$ Wb $\xi = \frac{N\Delta\Phi}{\Delta t} = \frac{0.03}{0.5}$ $= 0.06$ V (or 60 mV)	 1A 1M 1A	
				3	
	(b)	(i)	The change of (magnetic) flux linkage is double that in (a)(ii), i.e. 0.06 Wb.	1M/1A	
				1	
		(ii)	Direction of current : *PQRS*	1A	
				1	
	(c)	(i)		1A 1A	
				2	
		(ii)	Move / swing to the right initially / momentarily / briefly.	1A	
				1	

Solution	Marks	Remarks
10. (a) (i) $x = 3$	1A	
	1	
(ii) More neutrons are produced in each fission for triggering further fissions, i.e. $x > 1$.	1A	
	1	
(b) (i) $m = m_0 e^{-kt}$ $k = \frac{\ln 2}{t_{1/2}}$ $(= 9.846\times10^{-10}\ \text{yr}^{-1})$		
$0.06 = m_0\ e^{-\ln 2 \times \left[\frac{2\times10^9}{7.04\times10^8}\right]}$	1M	OR $0.06 = m_0\ (\frac{1}{2})^{2\times10^9/7.04\times10^8}$
$m_0 = 0.429882832\ \text{(kg)} \approx 0.430\ \text{(kg)}$	1A	
	2	
(ii) $\frac{0.430}{13.556 + 0.430} = 0.03073691 \approx 3.1\ \% > 3\%$ Thus natural nuclear fission was possible.	1M/1A	
	1	
(c) Underground water might run dry.	1A	
$\boxed{\text{OR Energy released by fission drys up the underground water.}}$	$\boxed{\text{1A}}$	
Therefore, fission might stop without slow neutrons.	1A	
	2	